Full Stack Assignment – 39

Assignment ID: FSD-39
Title: Delegation-Based Approval Management System
Duration: 3 Hours

1️⃣ Objective

Design and develop a **Delegation-Based Approval Management System** that allows users to submit approval requests while enabling managers to delegate approval authority to other users.
The system must enforce delegation rules and ensure **data persistence using a database**.

2️⃣ User Roles

REQUESTER

- Register and log in to the system
- Create approval requests
- View own request status

MANAGER

- Register and log in to the system
- Approve or reject requests
- Delegate approval authority to another user

DELEGATE

- Register and log in to the system
- Approve or reject delegated requests

3️⃣ Functional Requirements

Authentication & Authorization

- Implement user registration and login
- Use **JWT-based authentication**
- Protect all backend APIs using JWT
- Enforce **role-based access control** on APIs and frontend views

Request Management

- REQUESTER can submit approval requests with details
- Requests must be stored in the database
- MANAGER or DELEGATE can act on requests based on delegation status
- Request status must be retrieved dynamically

🔁 Delegation Management

- MANAGER can delegate approval authority to a DELEGATE
- Delegation details must be stored in the database
- Delegation can be time-bound or active/inactive (basic flag is enough)

4️⃣ Business Rule (Critical)

⚠️ A delegate can approve requests only while the delegation is active.

Approval attempts outside an active delegation window must be rejected.

5️⃣ Technical Constraints (Mandatory)

- All request and delegation data **must persist in a database**
- Hardcoded or in-memory storage is **not allowed**
- Backend APIs must reflect actual database state
- JWT authentication and role validation are mandatory
- Use environment variables for sensitive configuration

6️⃣ Frontend Requirements

- Login and registration pages
- Role-based dashboards for REQUESTER, MANAGER, and DELEGATE
- Request creation view for REQUESTER
- Delegation management view for MANAGER
- Approval action view for MANAGER / DELEGATE

UI design is not evaluated; delegation logic and correctness matter.

7️⃣ Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview
- Tech stack used
- User roles and permissions
- API endpoints
- Database schema or collections
- Steps to run the project locally
- Live frontend and backend links

8️⃣ Evaluation Note

Evaluation focuses on **delegation enforcement, role resolution, and end-to-end execution**.